

The importance of sensory feedback to enhance embodiment during virtual training of myoelectric prostheses users

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ABSTRACT

Amputation is a lasting life experience. Furthermore, prosthesis fitting and training are usually long and sometimes frustrating experiences. Few approaches explore the problem of controlling virtual myoelectric prostheses, since integration with current VR devices is not a trivial issue. Besides, the lack of proper sensory feedback often leads to an experience far too 'artificial' for most users. In this work, we propose a system that uses Immersive Virtual Reality (IVR) and EMG signal processing to provide a training environment for amputees who are supposed to use myoelectric prostheses. Here, we also investigate the efficiency of learning how to control a virtual prosthesis with and without sensory feedback. The system runs on a Vive Pro System using a Vive Tracker to track the user's arm position and orientation and mirror the movement into a virtual prosthesis. EMG signals control the opening and closing of a virtual prosthesis. Vibrational elements placed on the user's forearm provide sensory feedback.

Our results show that virtual training can be significantly improved when proper tactile feedback is provided, especially for myoelectric controlled system. EMG control and tactile feedback have provided an effective control loop, which allows for a better sense of realism for the training protocol.

Keywords: Immersive Virtual Reality, Amputee, Myoelectric prosthesis, Prosthetic training, sensory feedback.

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Youtube Link: <https://youtu.be/RYVbeHW0Kn8>



Figure 1: Virtual Reality environment.

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